



Capstone Project

Play Store App Review Analysis

By

Ajay Solanki

Data Science Trainee,



WHY ANALYZE THE GOOGLE PLAY STORE?



Mobile App Market
is set to grow 20%
by 2023



Android Apps
comprise 90% of the
Mobile App Market



What makes an App
popular? Can we predict
how popular it's going to
be?



What are some
interesting patterns in
user behavior related to
app usage & feedback?



Introduction

Play store is an Android Market serves as the official app store for certified devices running on the Android Operating system. Developed and Operated by Google, launched on 6th March, 2012.

Approximately 3.48 million apps are in the Play store. Play store apps have their own features such as Ratings, Reviews, Size and more. From the problem statement given, we should analyze the given database and should come up with the key factors that increased the number of users, long term usage etc., the objective of this project is to deliver insights to understand customer demands better and thus help developers to popularize the product.



Problem Statement

- ❑ Two datasets are provided, one with **basic information (play store Data.csv)** and the other with **user reviews (User Reviews.csv)** for the respective app.
- ❑ We must examine and evaluate the data in both datasets in order to identify the important characteristics that influence app engagement and success.

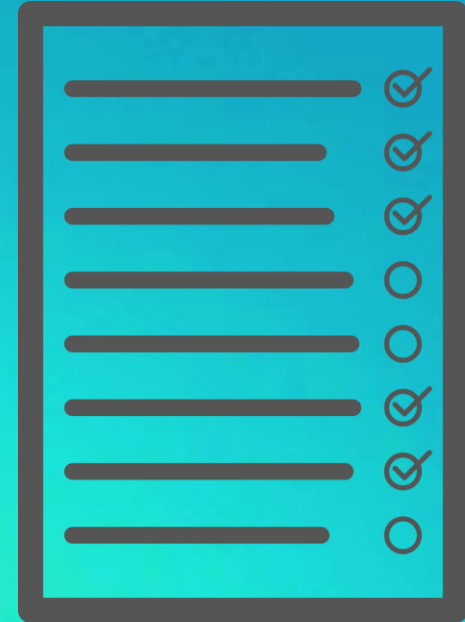
So, what factors influence an app's success

An app is said to be successful if it has:

- ❑ A high average user rating
- ❑ A good number of positive reviews
- ❑ A good number of monthly average users
- ❑ High revenue per customer and so on.



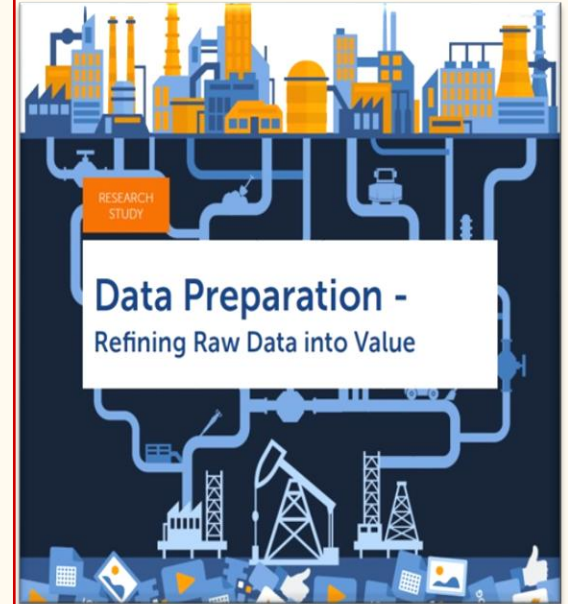
- ☐ Introduction
- ☐ Exploring the database
- ☐ Top 10 Highest rating Apps in google play store in terms of categories
- ☐ Number of Application in terms of Category
- ☐ Top 10 apps which has maximum downloads
- ☐ Which 10 apps from the 'FAMILY' category are having the lowest rating and highest rating.
- ☐ Free and Paid Apps
- ☐ Relation between app category and app price
- ☐ Filter out "junk" apps
- ☐ Sentiment analysis of user reviews
- ☐ conclusion





Dataset Preparation

- **Loading the data sets:** Two datasets, First Play store app dataset and User Reviews dataset.
- **Import Libraries:** NumPy, Pandas, Seaborn and Matplotlib
- **Data cleaning:** Null values, Finding and removing Outliers, Removing duplicate data.
- **Data Imputation:** Filling the missing categorical values with mode and numerical values with median. Conversion of price, installs, reviews into numerical values.
- **Exploratory Data Analysis:** Analyzing the data sets to summarize their main characteristics using statistical graphics and data visualizations method.





Attributes in Google Play store Data

- 1.App :** This column Contains the name of the app for each observation.
- 2.Category :** This column Contains Category to which the app belongs.
- 3.Rating :** This column contains the average rating for the app.
- 4.Reviews :** This column contains the number of reviews that the app has received on the play store.
- 5.Size :** This column contains the amount of memory the app occupies on the device.
- 6.Installs :** This column contains the number of times that the app has been downloaded and installed from the play store.
- 7.Type :** This column contains the information whether the app is free or paid.
- 8.Price:** If the app is a paid app, this column contains the data about its price.
- 9.Content Rating:** This column contains the maturity rating of the app i.e. the age group of the audience for which it is suitable.
- 10.Genres:** This column contains the data about to which genre the app belongs. Genres can be considered as a further division of the group of Category.
- 11.Last Updated:** Contains the date on which the latest update of the app was released.
- 12.Current Version:** Contains information on the current version of the app available on the play store.
- 13.Android Version:** Contains information about the android versions on which the app is supported.



Attributes in User reviews

1. **App-** Application name
2. **Translated Review-** User review
3. **Sentiment-** Positive/Negative/Neutral
4. **Sentiment Polarity-** Sentiment polarity score
5. **Sentiment Subjectivity-** Sentiment subjectivity score





OVERVIEW OF ANALYSIS

Data Cleaning



Understand the structure of the dataset and clean data before analysis

Data Exploration



Uncover initial patterns, characteristics, and points of interest using visual exploration

Predictive Modeling



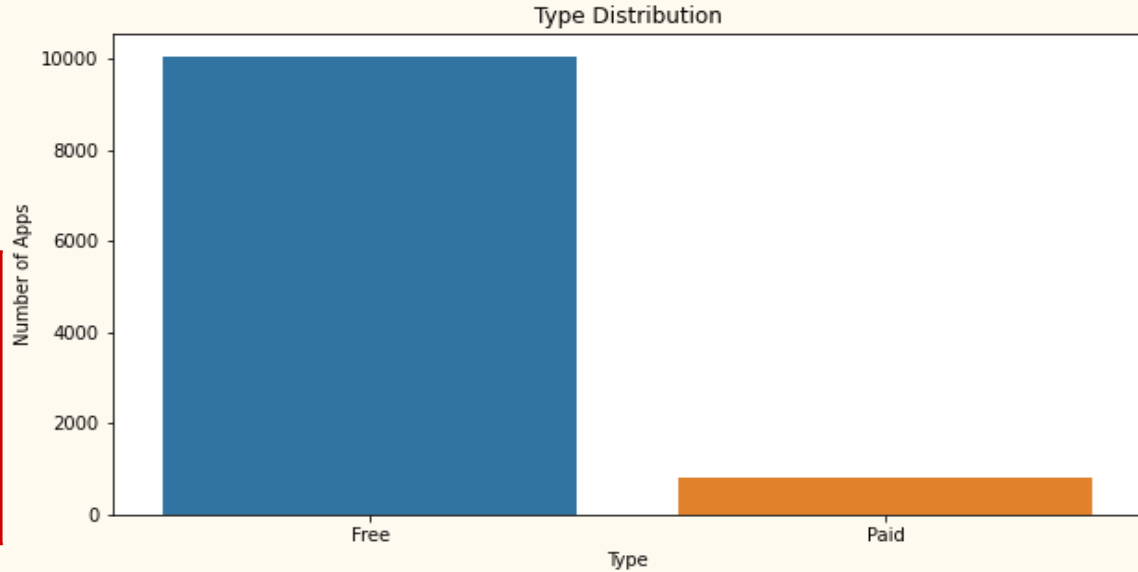
Formulate a statistical model to forecast an outcome using relevant predictors



Paid apps v/s Free apps

When comparing the both plots, people are showing more interest on free apps like art & design. And The number of free apps are more than 10000

When it's coming, commercial people are preferring apps like Business, communication and Personalization and the number of paid apps are less than free apps.





Top 10 Highest rating Apps in google play store in terms of categories

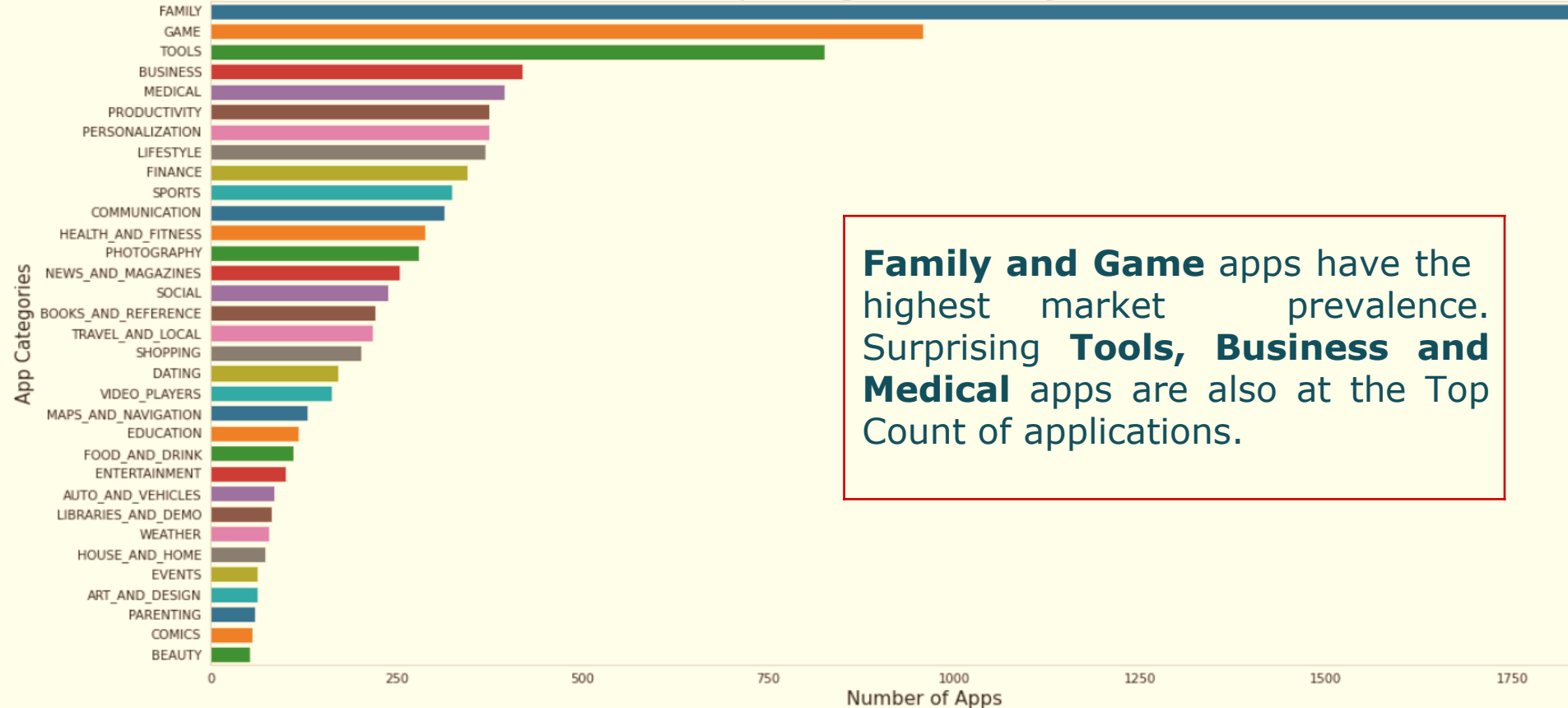
Categories like Games, Business, and Family are having comparatively more amount of apps count. We can see the data for our reference.

	App	Category	Rating
9056	Santa's Monster Shootout DX	Game	5.0
8395	DG TV	News_And_Magazines	5.0
8493	PK and DK Audio App	Family	5.0
6330	HON. B.J. ACS COLLEGE ALE	Family	5.0
6342	BJ Foods	Business	5.0
6363	Read it easy for BK	Lifestyle	5.0
9766	ER Assist	Productivity	5.0
6364	BK Video Status	Family	5.0
6372	BK Formula Calculator	Tools	5.0
6375	Dr Bk Sachin bhai	Lifestyle	5.0
9806	Dr. ES PV Calculator	Tools	5.0



Count of Applications in each category

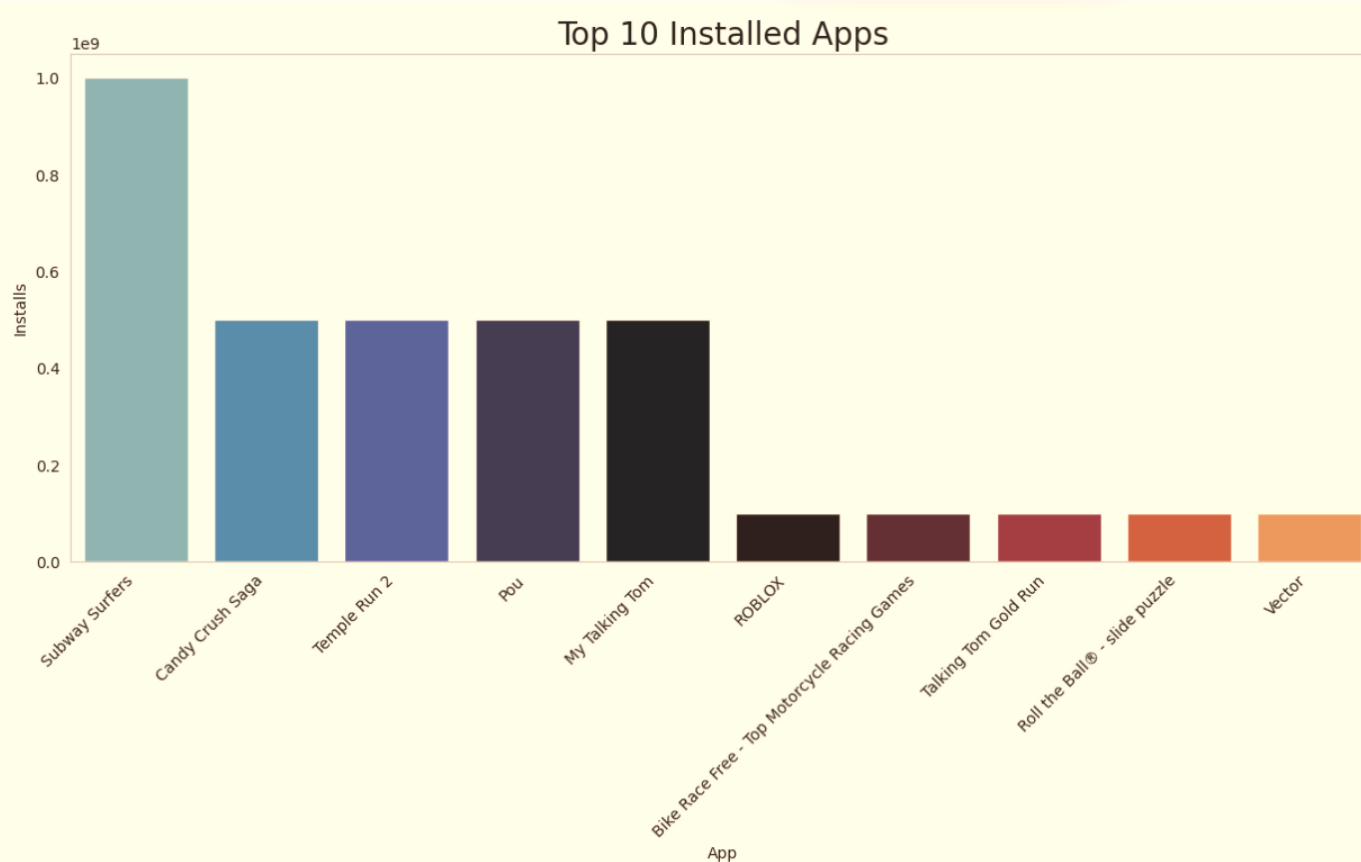
Top categories on Playstore



Family and Game apps have the highest market prevalence. Surprising **Tools, Business and Medical** apps are also at the Top Count of applications.



Top 10 installed apps in any category

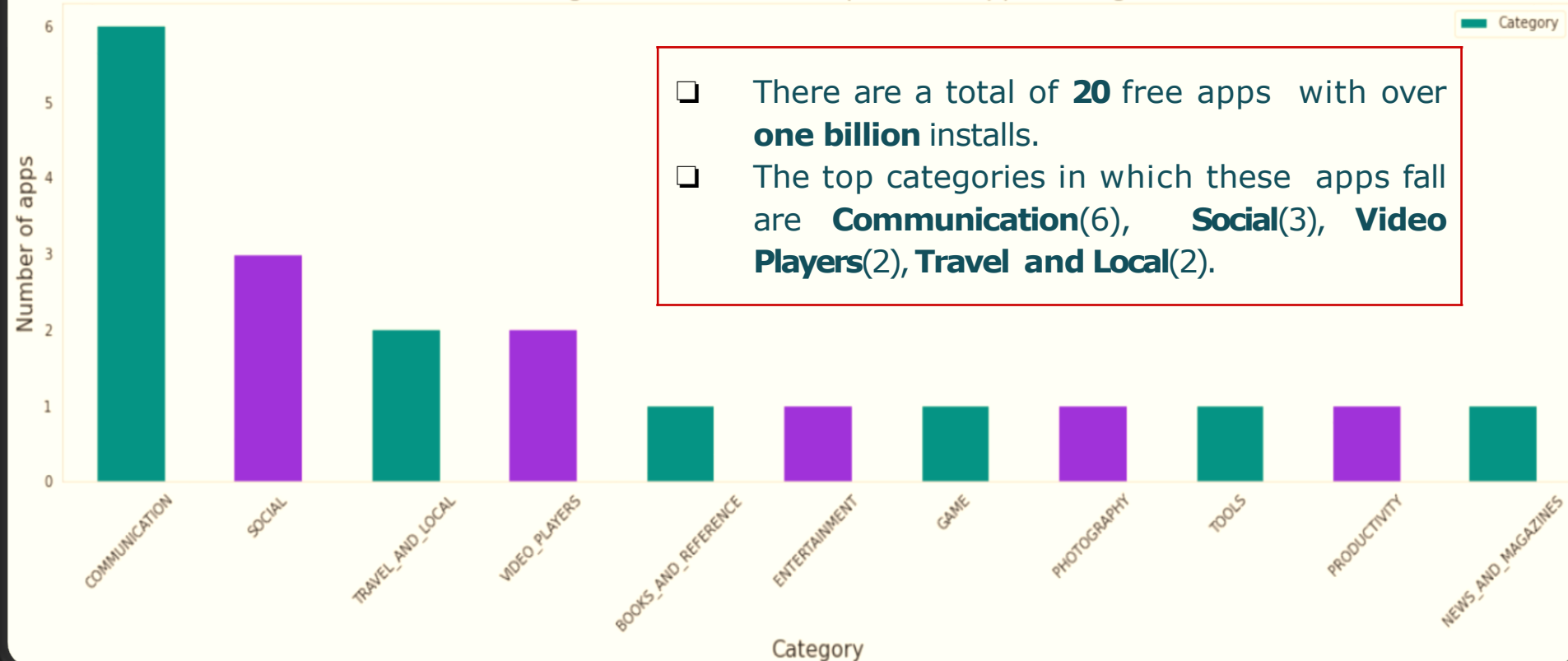


This graph shows the top installed apps in the '**Games**' category. Further looking into the play store reveals that these apps are light, casual, single player games.



Top Free Apps

Categories in which the top 20 free apps belong



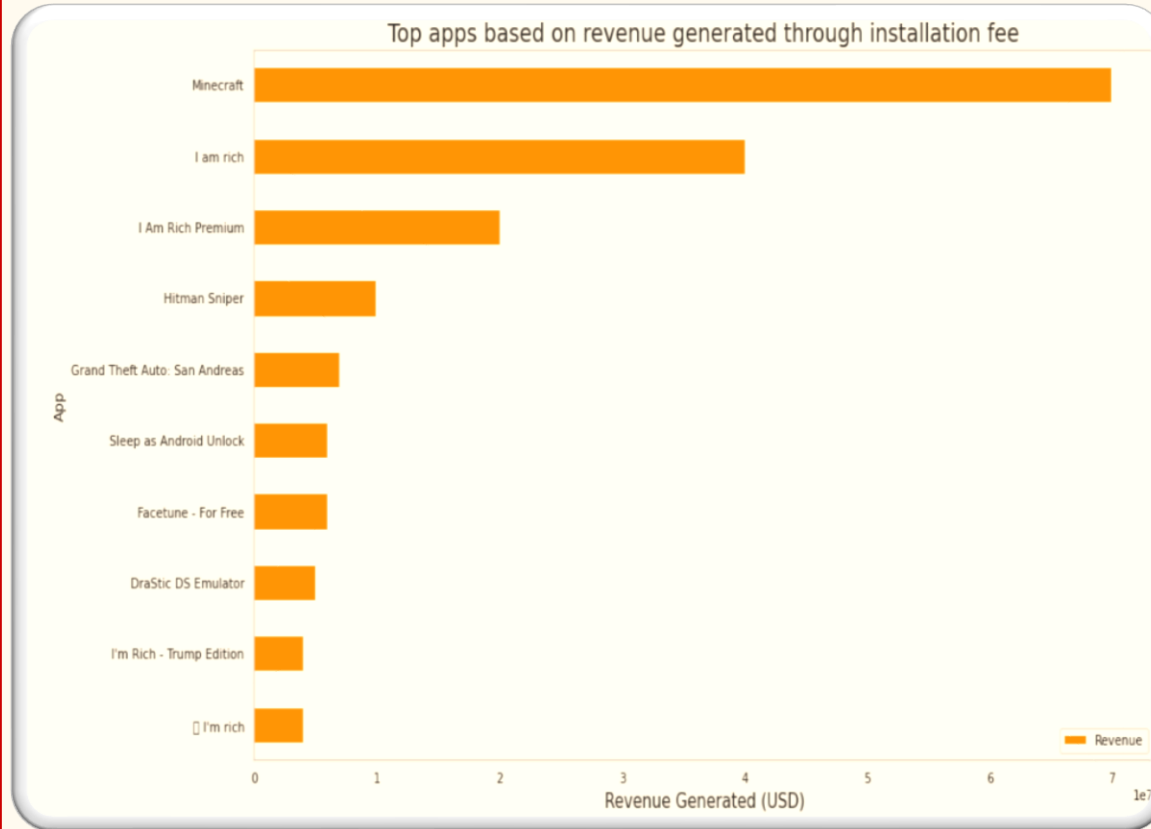


Top Paid Apps Based on RevenueGenerated

- Revenue generated is given by the formula:

$$\text{Revenue} = \text{Installs} * \text{Price}$$

- Note that in this case, revenue refers to the money earned only from paid app installs.
- The top categories in which these apps fall are **Lifestyle(5)**, **Family(5)**, and **Game(4)**.
- Minecraft**, **I am rich**, and **I am rich premium** are the top paid apps based on revenue generated.
- Minecraft** is the only app that has over **10 M** installs.

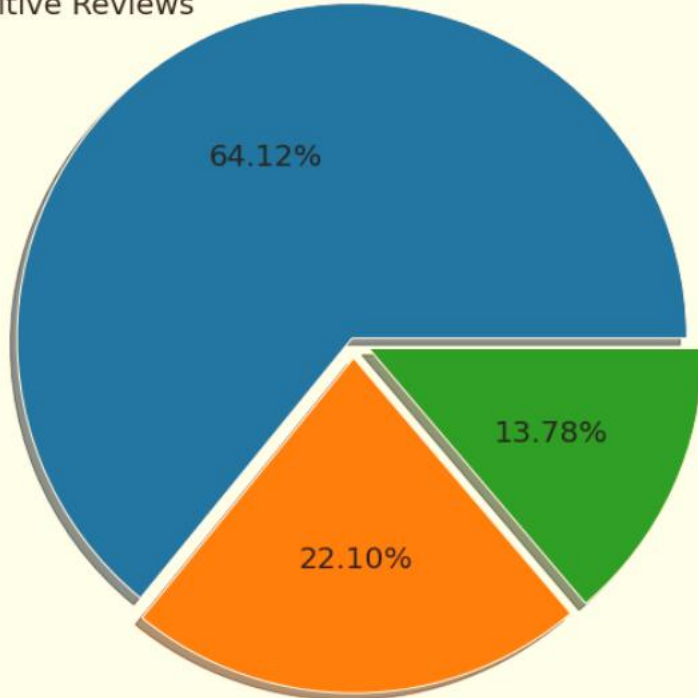




Percentage of Review Sentiments

Percentage of Review Sentiments

Positive Reviews



- Positive Reviews
- Negative Reviews
- Neutral Reviews

Neutral Reviews

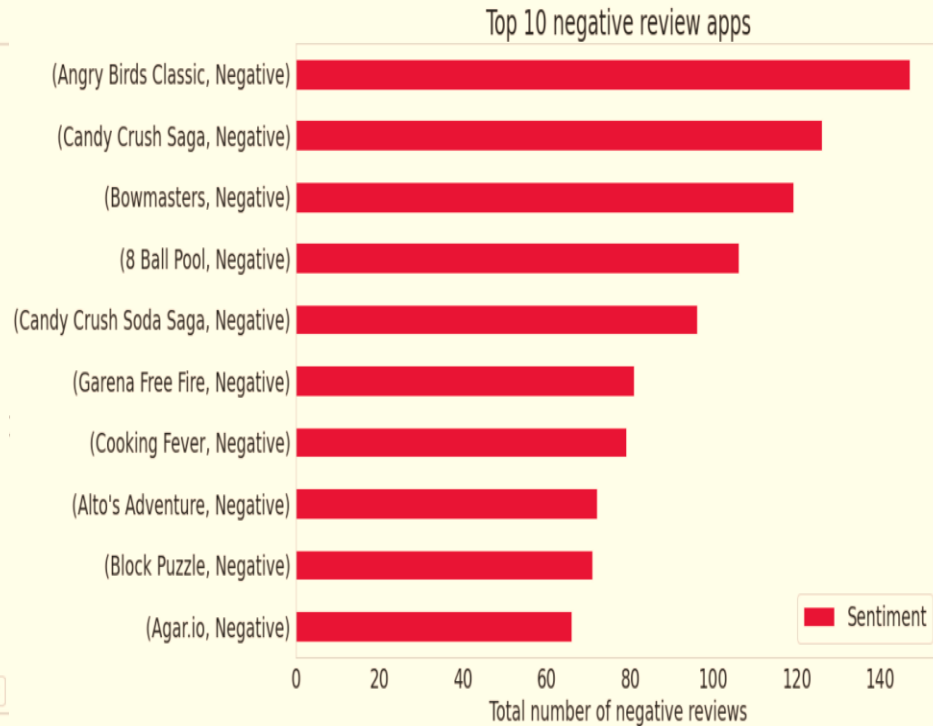
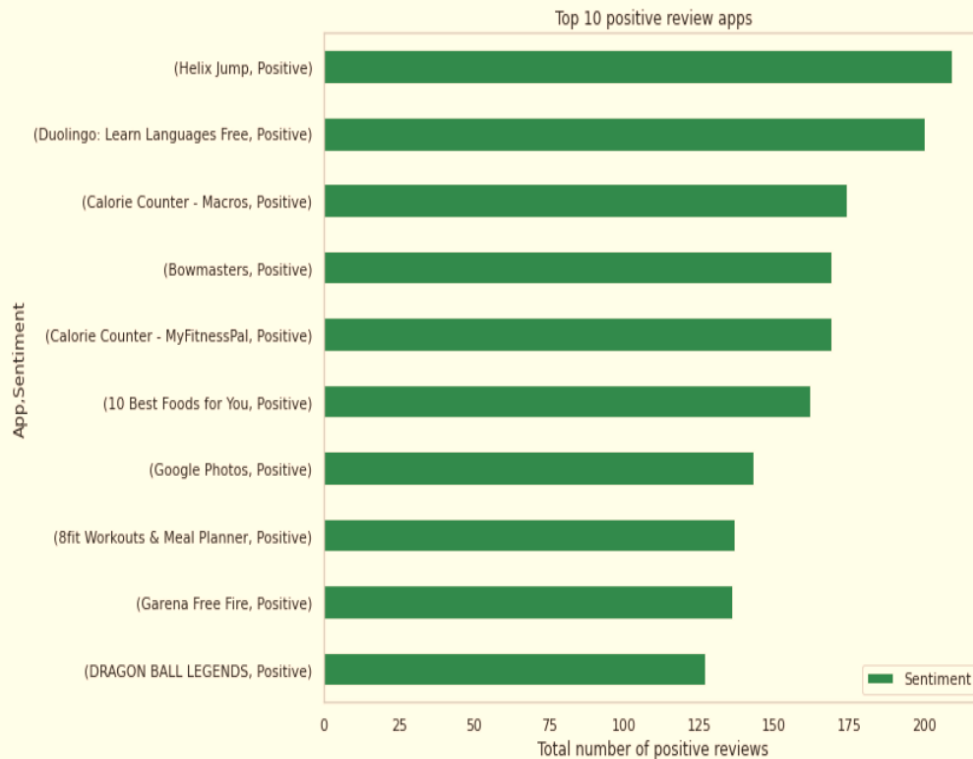
Negative Reviews

The number of **Unique** Apps from Play store and User reviews merged dataset are **816**.

From Sentiment column, **64%** are **Positive**, **22%** are **Negative** and **14%** are **Neutral** values.



Positive and Negative Reviews



Helix Jump is a App from merged dataset has highest **209 Positive** sentiment count.

Angry Bird Classic is a app from merged dataset has highest **147 Negative** sentiment count.

Challenges Faced

- ❑ Reading the dataset and comprehending the problem statement.
- ❑ Handling the error, duplicate and NaN values in the dataset.
- ❑ Designing multiple visualizations to summarize the information in the dataset and successfully communicate the results and trends to the reader.



Conclusion's

After analyzing the dataset we have got answers to some of the serious & interesting facts which any of the android users would love to know.

- a) Top 10 Highest rating Apps in google play store in terms of categories
- b) Number of Application in terms of Category
- c) Top 10 apps which has more downloads
- d) Which 10 apps from the 'FAMILY' category are having the lowest rating and highest rating.
- e) Free and Paid Apps
- f) Relation between app category and app price

